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Horn-VF.CONCEPT

SFRS for Zefiro-Europe

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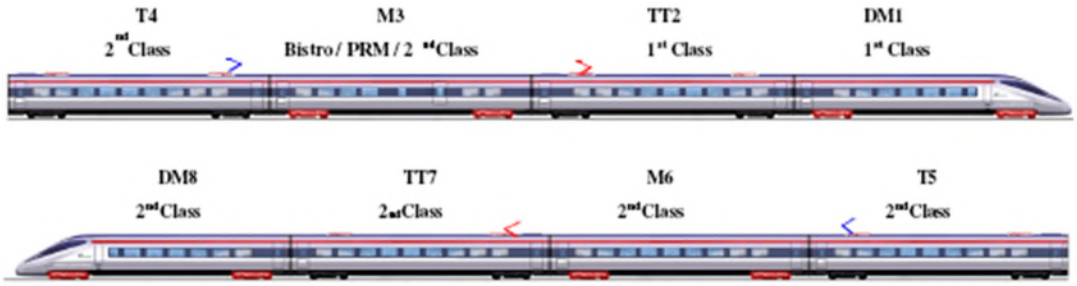
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1 Introduction

This document is a function specification for the control of the *Horn* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System

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2 System Overview

2.1 System Configuration

The macrophone is used by the driver to emit a warning signal. It is controlled via 2 switches placed on the drivers desk which work only the macrophones mounted at the attached front end. There is also one pneumatic emergency button that can be operated also when electricity is not available.

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3 Functional requirements

3.1 Function Horn

3.1.1 Function Design

Two pneumatic macrophones with two different pitches will be mounted in each front end of a train. The air supply will be controlled via magnetic valves which are operated by buttons placed on the drivers desk. The use of the macrophones shall be monitored by TCMS.
There will be one pneumatic emergency button that operates with 660 Hz. The emergency button will be possible to operate without electricity.

3.1.1.1 Normal condition

Macrophone system at the disposal of the driver to send out warning signals

3.1.1.2 Failure cases

A) One macrophone is at fault
B) Both macrophones are at fault
C) No electricity available on train

3.1.1.3 Failure detection

A) Detected during routine test carried out when the train is powered up.
B) Detected during routine test carried out when the train is powered up.
C) Detected by the driver

3.1.1.4 Failure handling

A) Train available for unlimited service. Repair during next depot visit.
B) Train out of service. Repair during corrective maintenance.
C) The driver can use the emergency button for macrophone.

3.1.2 Requirements

ID	2F_02.07.Zefiro-Europe.TRS.13
Description	The horn shall be pneumatically operated.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.07.Zefiro-Europe.CONCEPT.53	The horn shall be pneumatically operated	Derived Requirement	4S_02.01.05.-.Horns and Macrofons		N/A
2F_02.07.Zefiro-Europe.CONCEPT.69	For each horn the air supply shall be controlled by a separate magnetic valve.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.07.Zefiro-Europe.CONCEPT.87	The high frequency horn shall be additionaly controlled via a pure pneumatic system connected to an emergency button	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.07.Zefiro-Europe.CONCEPT.70	Each horn shall be separatly blockable by a cut-off cock.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

ID	2F_02.07.Zefiro-Europe.TRS.11
Description	The horn shall be activated by a control in the driver´s desk.
Safety level	SIL 0

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ID	2F_02.07.Zefiro-Europe.TRS.15
Description	There shall be two controls in the cab to operate the horn. One control shall be for the driver to operate the horn and one control shall be for the second driver to operate the horn.ko
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.07.Zefiro-Europe.CONCEPT.54	2 three-position-switches (one for driver one for second driver) shall be positioned on the drivers desk with the functions – low frequency horn(unstable), – horn off (stable) – high frequency horn(unstable)	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_02.07.Zefiro-Europe.CONCEPT.79	A pneumatic emergency button shall be installed on the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

The emergency push button shall be connected to the high frequency horn.

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.07.Zefiro-Europe.CONCEPT.35	The magnetic valve of the high frequency horn shall be connected to the switches on the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.07.Zefiro-Europe.CONCEPT.36	The magnetic valve of the low frequency horn shall be connected to the switches on the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.07.Zefiro-Europe.CONCEPT.71	The activation of the horns shall be monitored by the juridical recorder. (Electrical switches and emergency button)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.07.Zefiro-Europe.CONCEPT.92	Horn control elements (Electrical switches and emergency button) shall have an electrical link to the juridical recorder.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.07.Zefiro-Europe.CONCEPT.74	The horn control switches shall be active when battery power is available	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_02.07.Zefiro-Europe.TRS.14
Description	When activating the emergency stop the horn shall automatically sound.
Safety level	SIL 0

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.28	n/a		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.29	n/a		

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.30	Switches or pneumatic emergency button for horn activation used.	L	2F_02.07.Zefiro-Europe.CONCEPT.71

Outputs:

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.32	n/a		

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_02.07.Zefiro-Europe.CONCEPT.33	n/a		

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5 Revision Log

5.1 Revision log 3EST000225-7905_A

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Date:	**2018-05-15**		
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Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_02.07.ISSUES.-.Zefiro-Europe.4		Remove supervision of cut out cocks by TCMS	Remove supervision of cut out cocks by TCMS Removed requirement: 2F_02.07.Zefiro-Europe.CONCEPT.75 2F_02.07.Zefiro-Europe.CONCEPT.76 2F_02.07.Zefiro-Europe.CONCEPT.88 Removed signal 2F_02.07.Zefiro-Europe.CONCEPT.89

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